
Where Does Platonic Convergence Live? A Linear Readout Gains Where the Galaxy Embedding Bends Toward the Label and Loses Where It Bends Away

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Abstract

1 The Platonic Universe tested the Platonic Representation Hypothesis (PRH) by
2 measuring the convergence of foundation-model representations of galaxy images
3 with mutual k -nearest-neighbour agreement (MKNN). We ask whether that con-
4 vergence is concentrated where the embedding manifold is flat. A pre-registered
5 raw-decoder mean-curvature statistic is negatively associated with cross-survey
6 agreement, but density removes most of the signal and the one residual case disap-
7 pears once its radial component is removed. A second-order expansion of a linear
8 map restricted to the manifold explains why: a readout sees $\langle w_N, \Pi \rangle$, the second
9 fundamental form contracted with its normal component; an alignment's second-
10 order error is a connection mismatch plus a relative extrinsic term; mean curvature
11 is only the metric trace of Π and determines neither. On 86,471 labelled galaxies
12 a linear readout gains accuracy where the manifold bends toward the label's own
13 curvature and loses it where the manifold bends away. In a counterfactual local
14 second-order surrogate of the readout, the manifold's bending in the readout's nor-
15 mal direction raises local R^2 at 79–99% of anchors, its sign reversal lowers it at
16 97–100%, and a random orientation of matched strength shows no such asymme-
17 try and is typically harmful under either sign, on five encoders; across anchors,
18 the mismatch between the label Hessian and the probe-facing Hessian is negative
19 wherever the label has local structure and survives cross-fitting and a change of en-
20 coder. Mean curvature of a single representation does not explain weak Platonic
21 convergence; the obstruction is relative and, for the fitted global linear alignment,
22 present at first order.

23 1 Introduction

24 The Platonic Representation Hypothesis [Huh et al., 2024] holds that models trained on different
25 data converge toward a shared representation; Duraphe et al. [Duraphe et al., 2025] tested it on
26 galaxy images: galaxies imaged by two surveys (Legacy Survey and Hyper Suprime-Cam, HSC)
27 share only 0.3 to 2.3 percent of nearest neighbours across surveys under a family of vision models.
28 We ask where on the manifold that agreement is concentrated: if two models disagree most where
29 the embedding surface bends sharply, local mean curvature should associate negatively with local
30 cross-survey agreement.

31 We contribute (i) a pointwise mean-curvature and second-fundamental-form instrument from a
32 trained decoder, validated against analytic truth; (ii) a density-controlled test of the curvature–
33 alignment prediction, which finds no effect free of density; (iii) the residual expansion of a linear map
34 on a submanifold, which names the curvature such a map responds to; and (iv) known-surface, cross-
35 encoder and interventional tests on the readout side: a linear readout gains local accuracy where the

manifold bends toward the label’s curvature and loses it where the manifold bends away, at nearly every anchor on five encoders.

2 Pointwise curvature from a decoder

Data. DINOv3 [Siméoni et al., 2025] ViT-B/16 embeddings of 101,725 galaxies imaged by both HSC [Aihara et al., 2018] and the Legacy Survey [Dey et al., 2019] (Platonic Universe release [Duraphe et al., 2025]); we draw 10,000 rows once with a recorded seed. A second set of 86,471 single-survey galaxies embedded by a supervised ImageNet-21k ViT-B/16 (“ViT-B” below; not DINOv3) with catalogue labels (r -band magnitude, photometric redshift, smooth fraction, stellar mass; the AstroPT galaxy set [Smith et al., 2024]) serves the readout experiment. We unit-normalize embeddings (data on S^{767}).

Instrument. Let $F : \mathbb{R}^d \rightarrow \mathbb{R}^{768}$ be the decoder of a plain auto-encoder (three hidden layers of width 250, SiLU), fit on 80% of rows; we evaluate curvature on held-out rows. With $J = DF(z)$, metric $g = J^\top J$, normal projector $P_N = I - Jg^{-1}J^\top$ and second fundamental form $\Pi = P_N D^2 F(z)$, the mean-curvature vector of the raw decoder surface is $H_{\text{raw}} = \text{tr}_g \Pi$, by automatic differentiation. On the unit sphere every d -submanifold has radial mean curvature $-d$, which says nothing about shape; differentiating $F/\|F\|$ instead splits H exactly as $-d\hat{x} + H^S$ ($H_{\text{rad}} = -d$ to 10^{-14}), and H^S carries the shape. Section 3 uses the pre-registered $\|H_{\text{raw}}\|$; Sections 4–5 use the projected decoder. Decoder curvature has been a regularizer [Lee and Park, 2023] and an estimator [Acosta et al., 2023].

Validation. Against closed-form curvature, at $D = 768$ an in-sphere surface $G(z) = Q \text{normalize}([\phi(z); 0.8h(z); 0 \dots])$ (ϕ inverse stereographic $\mathbb{R}^{16} \rightarrow S^{16}$, h a scalar sum of four Gaussian bumps, Q a rotation; 18-dimensional span) at $n = 86,471$, $k = 2,048$ gives rank 0.94 against the true $\|H^S\|$, direction cosine 0.999 and magnitude ratio 1.00 for the full vector.

3 Curvature and cross-survey convergence

We correlate the per-point mutual k -nearest-neighbour score $\text{MKNN}_i = |N_k(z_1)_i \cap N_k(z_2)_i|/k$ at $k = 20$ between the HSC and Legacy Survey embeddings with $\|H_{\text{raw}}\|$ at $d \in \{20, 25, 32\}$. The raw association $\rho(\|H_{\text{raw}}\|, \text{MKNN})$ is -0.112 , -0.128 and -0.024 at $d = 20, 25, 32$ ($p < 10^{-5}$, $< 10^{-5}$, 0.009). But k -NN statistics are density-sensitive [Radovanović et al., 2010]: local density ($k = 30$ estimate on the Legacy embedding) correlates with agreement at -0.21 at every d and with $\|H_{\text{raw}}\|$ at $+0.43$, $+0.32$ and $+0.01$, so residualizing on density leaves -0.024 , -0.066 and -0.022 . Under a permutation null within density strata only $d = 25$ survives ($p < 5 \times 10^{-5}$; CKA [Kornblith et al., 2019] agrees), and removing the radial component of H_{raw} collapses it to -0.023 .

4 What a linear map sees on a curved manifold

Let $M \subset \mathbb{R}^D$ be a d -dimensional submanifold, $d < D$; at x_0 take an orthonormal tangent frame J , the second fundamental form Π (normal-valued, symmetric $d \times d$), $H = \text{tr} \Pi$, and for a vector w its tangent and normal parts $w_T + w_N$. In normal coordinates u at x_0 , M is $x_0 + Ju + \frac{1}{2}\Pi(u, u) + O(\|u\|^3)$ [Lee, 2018], so

$$[\nabla_M(w \cdot x)]_J = J^\top w, \quad \text{Hess}_M(w \cdot x) = \langle w_N, \Pi \rangle, \quad \Delta_M(w \cdot x) = \langle w, H \rangle. \quad (1)$$

For $M \subset S^{D-1}$, $\Pi = \Pi^S - g\hat{x}$ splits off the sphere’s radial direction, so $\text{Hess}_M(w \cdot x) = \langle w_N, \Pi^S \rangle - (w \cdot \hat{x})g$ (2): a linear readout of a unit-normalized embedding is bent by the manifold’s shape and, whatever the manifold, by the sphere in proportion to the readout’s own value $\hat{y} - b_0$. For a label y on M , a global affine probe $\hat{y} = w \cdot x + b_0$ and a patch $u \sim N(0, s^2 I_d)$ in normal coordinates, embedding and label locally C^4 (Gaussian constant $\frac{1}{2}$; real k -NN patches share only the tensor structure), with $\bar{c}$ the patch-mean residual, $\delta = [\nabla_M y]_J - J^\top w$, $\Delta = \text{Hess}_M y - \langle w_N, \Pi \rangle$,

$$\mathbb{E}_{\text{patch}}[(y - \hat{y})^2] = \bar{c}^2 + s^2 \|\delta\|^2 + \frac{1}{2}s^4 \|\Delta\|_F^2 + O(s^4 \|\delta\|) + O(s^6). \quad (3)$$

For two embeddings F, G of the same objects in a common chart u and a linear map A with $AJ_F = J_G$, the same expansion gives $G - (G_0 + A(F - F_0)) = \frac{1}{2}[J_G(\Gamma^G - \Gamma^F) + \Pi_G - A\Pi_F](u, u) +$

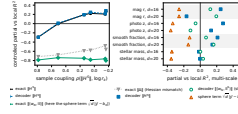


Figure 1: (a) Known surface, linear label, exact geometry: three-control partial vs local R^2 as the density-curvature coupling is set by construction (γ from -1 to 1). (b) Galaxies, 512 anchors, $k = 2,048$, multi-scale control: the decoder’s $\|H^S\|$ (filled squares), shape term (open circles), sphere term (open triangles).

label	$\ H^S\ $		shape $\ \langle w_N, \Pi^S \rangle\ $		sphere $\sqrt{d} \hat{y} - b_0 $		mismatch $\ \Delta\ $		alignment	
	$d=16$	$d=20$	16	20	16	20	16	20	16	20
mag_r	+0.25	-0.17	+0.12	+0.19	-0.26	-0.26	-0.39	-0.40	+0.35	+0.43
photo_z	+0.28	+0.13	-0.06*	-0.12	-0.33	-0.36	-0.45	-0.44	+0.26	+0.24
smooth_fraction	+0.21	+0.15	+0.07*	+0.16	-0.13	-0.15	-0.09*	-0.17	+0.19	+0.07*
stellar_mass	-0.02*	-0.00*	+0.00*	+0.03*	-0.35	-0.35	-0.04*	-0.09*	+0.01*	+0.02*

Table 1: Galaxies, 512 anchors, $k = 2,048$: rank-partial Spearman against local R^2 under the multi-scale density control. * not significant at 0.05.

81 $O(|u|^3)$, with Γ the induced connections: the second-order residual contains a connection mismatch
82 and a *relative* extrinsic term, whose target-normal component is $\Pi_G - P_N^G A \Pi_F$.
83 So $\|H\|$, the metric trace’s norm, need not correlate with either term; and since the second-order
84 term scales as s^4 and fixed- k neighbourhoods let density set s , patch-scored curvature effects are
85 density-sensitive.

86 5 Probe-facing curvature: bending toward the label helps, bending away 87 hurts

88 Statistic: five-fold out-of-fold ridge probe [Alain and Bengio, 2016] at the pre-registered $\alpha = 100$
89 from embedding to label; local R^2 inside each of 512 anchors’ 2,048-neighbourhoods; rank-partial
90 Spearman against local R^2 with Freedman–Lane p -values [Freedman and Lane, 1983], control-
91 ling for log patch radius, local label variance and count and, stronger, log radius at five scales
92 $k \in \{16, \dots, 2048\}$. Per anchor we compute $\|H^S\|$ and $\|\langle w_N, \Pi \rangle\|_g$, split by Eq. (2) into shape
93 and sphere terms.

94 **Known surface.** Resampling the validation surface with weight $\|H^S\|^\gamma$ (Figure 1a) moves the
95 $\|H^S\|$ partial from -0.29 to $+0.26$, reproduced by the decoder, while $\|\langle w_N, \Pi \rangle\|$ holds at -0.74 to
96 -0.80 , there dominated by the sphere term of Eq. (2).

97 **Galaxies.** Here the two terms of Eq. (2) trade places: the shape term dominates the probe-facing
98 Hessian (rank 0.96–0.99 with the full norm). We estimate $\text{Hess}_M y$ per anchor by a least-squares
99 quadratic of the label on the tangent-projected coordinates $u = g^{-1} J^\top (x - x_0)$ of its 2,048 neigh-
100 bours (a Monge chart, so the coefficient is the covariant Hessian). Three regularities (Table 1; Ap-
101 pendix A). The sphere term is negative on every label and both d (-0.13 to -0.36) under the sealed
102 probe; with a weak-ridge probe ($\alpha = 1$, Appendix B) it stays negative for magnitude and smooth
103 fraction (-0.21 , -0.12) and vanishes for redshift and stellar mass (ridge shrinkage of extreme pre-
104 dictions). Wherever the label has local structure (magnitude, redshift, smooth fraction; stellar mass
105 is null throughout) the mismatch is negative (-0.09 to -0.45 ; unchanged under cross-fitting, Ap-
106 pendix B) and the alignment $\cos_g(\text{Hess}_M y, \langle w_N, \Pi \rangle)$ positive ($+0.07$ to $+0.43$). The shape term
107 alone is label-signed (positive for magnitude and smooth fraction, null to weakly negative for red-
108 shift): with $K = \langle w_N, \Pi \rangle$, $\|\Delta\|^2 = \|\text{Hess}_M y\|^2 - 2\langle \text{Hess}_M y, K \rangle + \|K\|^2$, so the cross term
109 decides for infinitesimal scaling from zero and $\|K\|^2$ is the finite-magnitude overshoot penalty. The
110 same regression on the readout $w \cdot x$ returns the point cloud’s own $\langle w_N, \Pi \rangle$ (median cosine with the
111 decoder’s 0.55–0.65).

Bending toward helps, bending away hurts. By Eq. (3) curvature lowers the second-order residual relative to $K = 0$ iff $2\langle \text{Hess}_M y, K \rangle > \|K\|^2$: the manifold must bend toward the label’s own curvature, and not too far. (Liu et al. [2023] derive the optimal normal weight of least squares on a hypersurface as a uniqueness and stability condition; manifold capacity [Chung et al., 2016, Slatton et al., 2026] uses dimension and radius, flattening [Psenka et al., 2024] treats curvature as an obstacle, latent curvature tracks generalization [Kaufman and Azencot, 2023] and biases local regression [Cheng and Wu, 2013]; none conditions on the sign of $\langle \text{Hess}_M y, K \rangle$.) The partials compare anchors, and no anchor is flat, so we test the condition with a counterfactual local second-order surrogate of the readout. Writing $K = K_S - (w \cdot \hat{x})g$ with $K_S = \langle w_S, \Pi^S \rangle$ the in-sphere shape term, each anchor’s 2,048 neighbours are scored by $\hat{y}_t(u) = c + a^\top u + \frac{1}{2}K_{\text{sph}}(u, u) + \frac{t}{2}K_S(u, u)$ with a the readout’s tangent gradient, K_S from the decoder and c refit: $t = 1$ is the manifold’s bending as seen in the readout’s normal direction (“bending toward the label” means $\langle R, K_S \rangle > 0$ for the remaining label curvature $R = \text{Hess}_M y - K_{\text{sph}}$), $t = -1$ its sign reversal, $t = 0$ shape-flat. In the Gaussian model $t = 1$ beats shape-flat iff $2\langle R, K_S \rangle > \|K_S\|^2$; on the actual neighbours, with r_0 the centred shape-flat residual and q_c the centred quadratic $\frac{1}{2}K_S(u_i, u_i)$, the exact criterion is $2\langle r_0, q_c \rangle > \|q_c\|^2$ with optimum $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ (Appendix E). On five encoders and all four labels, $t = 1$ beats shape-flat at 79–99% of anchors and $t = -1$ is worse at 97–100%, while a random orientation with the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign ($t = 1$ beats shape-flat at 22–36% of anchors, $t^* \approx 0$; Appendix E). The probe-selected shape quadratic is locally error-reducing; under the second-order model that is the signature of bending toward the remaining label curvature. This within-anchor question differs from Table 1’s cross-anchor one (does variation in mismatch explain variation in local R^2) and can hold where that has no signal, as for stellar mass. t^* sits at 1.5 to 3.6 throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial second-order term.

Across encoders. The quadratic-chart estimator’s mean-bending trace at the same anchors of five encoders (the readout set’s supervised ViT-B, DINOv3 ViT-B/16, CLIP-B, ConvNeXt-B, ViT-L; $d = 16$) has density-controlled associations with the magnitude probe’s local R^2 of both signs (-0.24 to $+0.27$), as Eq. (1) allows. The decoder test repeated per encoder (Appendix D) is encoder-stable where the expansion says it should be: the mismatch partial is negative for magnitude and redshift on all five encoders (-0.32 to -0.61 ; cross-fitted -0.28 to -0.62) and the alignment positive for magnitude on all five ($+0.13$ to $+0.53$) and for redshift on four; the shape term’s sign varies with encoder as well as label; stellar mass stays null but for one mismatch cell (-0.12).

6 Discussion

A linear readout of a galaxy embedding gains accuracy where the manifold, seen in the readout’s normal direction, bends toward the label’s curvature and loses it where that bending is reversed, at nearly every anchor and on five encoders. The residual expansion names the quantity, $\langle w_N, \Pi \rangle$ against $\text{Hess}_M y$, and a decoder’s pointwise second fundamental form measures it at production scale, validated against analytic truth for H^S and in substantial agreement with the data for the contracted Hessian. For a linear alignment map, the mechanism recent accounts of alignment assume [Bangachev et al., 2026], the relevant second-order extrinsic normal term is $\Pi_G - P_N^G A \Pi_F$, not one embedding’s mean curvature; the full residual also carries the connection mismatch; our pilot (Appendix C) finds a 44% first-order tangent-frame residual and no estimator-stable second-order association. On this survey pair, single-model mean curvature has no robust density-controlled association with cross-survey convergence.

Limitations. One survey pair. The label’s Hessian is a local estimate (R^2 gain over linear 0.10–0.17); its norm is reliably ranked (split-half ρ 0.91–0.94) but its direction only moderately (tensor cosine 0.19–0.35). On a synthetic surface with known curvature the mismatch partial reproduces (-0.14 to -0.88) but the alignment partial has no stable sign; the intervention of Appendix E carries the alignment claim. Mismatch and alignment are the primary hypotheses, the rest exploratory. Permutation p -values assume anchor exchangeability under overlapping patches.

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Appendix A: decoder ablations

Table 2 repeats the four quantities of Table 1 at $d = 16$ under the multi-scale density control for two further decoder initialisation seeds and for a decoder of width 400^3 instead of 250^3 (all fits reach variance explained 0.952). Every significant sign of the main table is retained.

label	quantity	seed 0	seed 1	seed 2	400^3
mag_r	shape	+0.12	+0.05*	+0.12	+0.05*
	sphere	-0.26	-0.26	-0.27	-0.26
	mismatch	-0.39	-0.35	-0.34	-0.34
	alignment	+0.35	+0.31	+0.37	+0.41
photo_z	shape	-0.06*	-0.09*	-0.09	-0.06*
	sphere	-0.33	-0.35	-0.34	-0.35
	mismatch	-0.45	-0.43	-0.42	-0.41
	alignment	+0.26	+0.24	+0.25	+0.25
smooth_fraction	shape	+0.07*	+0.20	+0.15	+0.10
	sphere	-0.13	-0.15	-0.14	-0.11
	mismatch	-0.09*	-0.05*	-0.08*	-0.07*
	alignment	+0.19	+0.14	+0.09	+0.10
stellar_mass	shape	+0.00*	-0.06*	+0.05*	+0.09
	sphere	-0.35	-0.35	-0.35	-0.35
	mismatch	-0.04*	+0.03*	+0.02*	+0.03*
	alignment	+0.01*	-0.02*	+0.03*	-0.02*

Table 2: Decoder ablations, $d = 16$, 512 anchors, multi-scale density control. * not significant at 0.05.

Appendix B: sensitivity of the mismatch and sphere columns

Cross-fitting: each 2,048-patch is split at random into halves; $\text{Hess}_M y$ is fitted on one half and local R^2 scored on the other (both directions shown), so the Hessian estimate and the outcome share no rows. The split-half cosine between the two Hessian estimates is the reliability of the estimate. Weak ridge: the probe refit at $\alpha = 1$ instead of 100 (global out-of-sample R^2 rises by 0.10 to 0.15), which substantially reduces the shrinkage of extreme predictions; the sphere term is recomputed from that probe.

label	d	mismatch, cross-fit		alignment, cross-fit		Hess. split cos	sphere, $\alpha = 1$
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	p50	(main: $\alpha = 100$)
mag_r	16	-0.38	-0.34	+0.33	+0.31	+0.35	-0.21 (-0.26)
mag_r	20	-0.43	-0.40	+0.39	+0.37	+0.24	-0.22 (-0.26)
photo_z	16	-0.42	-0.36	+0.23	+0.20	+0.32	+0.05* (-0.33)
photo_z	20	-0.42	-0.42	+0.22	+0.22	+0.20	+0.03* (-0.36)
smooth_fraction	16	-0.10	-0.17	+0.17	+0.19	+0.33	-0.12 (-0.13)
smooth_fraction	20	-0.15	-0.21	+0.05*	+0.03*	+0.23	-0.13 (-0.15)
stellar_mass	16	+0.01*	-0.04*	-0.03*	+0.06*	+0.31	-0.02* (-0.35)
stellar_mass	20	-0.06*	-0.07*	+0.01*	-0.00*	+0.19	-0.02* (-0.35)

Table 3: Cross-fitted mismatch and alignment partials (multi-scale control), split-half reliability of the label Hessian, and the sphere-term partial under a weak-ridge probe.

Appendix C: relative second fundamental form, pilot

Two sphere-projected decoders (HSC = F , Legacy = G ; Phase 7 protocol), a global ridge map A from x_F to x_G (fit on the 8,000 training rows), 2,048 seeded anchors. Columns: holdout R^2 of A ; median first-order obstruction $\|AJ_F - J_GL\|/\|AJ_F\|$ with $L = J_G^+ AJ_F$; then multi-scale density-controlled partials against MKNN ($k = 20$; log radius at $k \in \{10, 30, 100, 300\}$ in both spaces) of each decoder's $\|H^S\|$, the first-order obstruction, $\|\Pi_G(L, L) - P_N^G A \Pi_F\|$, the same with L fitted

on 256 neighbours' latent codes, and a decoder-free estimate from the quadratic coefficient of the alignment residual on F 's tangent coordinates.

run	R_A^2	tan. resid	$\ H^S\ $ HSC	$\ H^S\ $ Legacy	tan. resid	Π_{rel}	Π_{rel} (local L)	Π_{rel} (data)
$d = 20$, seed 0	0.64	0.44	-0.15	+0.04*	-0.18	+0.04*	-0.18	+0.13
$d = 25$, seed 0	0.64	0.43	-0.10	+0.06	-0.16	+0.05	-0.23	+0.13
$d = 20$, seed 1	0.64	0.44	-0.11	+0.05	-0.19	+0.04	-0.20	+0.14

Table 4: Relative-II pilot. * not significant at 0.05.

Appendix D: the same test on four further encoders

Table 5 repeats Table 1 at $d = 16$ on the same 86,471 galaxies embedded by four further encoders from the same release (DINOv3 ViT-B/16, CLIP ViT-B, ConvNeXt-B, ViT-L; widths 768, 512, 1,024, 1,024; each unit-normalized), with a decoder of the same width and protocol fitted per encoder (seed 0; variance explained 0.966, 0.985, 0.965, 0.968 for the four new fits), 512 anchors, $k = 2,048$, multi-scale density control. The first row per label is the probe's global out-of-sample R^2 ; Table 6 gives the cross-fitted mismatch and alignment and the split-half reliability of the label Hessian per encoder.

label	quantity	ViT-B	DINOv3	CLIP-B	ConvNeXt-B	ViT-L
mag_r	global R^2	0.52	0.57	0.50	0.52	0.52
	shape	+0.12	-0.11	+0.16	+0.16	-0.21
	sphere	-0.26	-0.23	-0.40	-0.24	-0.08*
	mismatch	-0.39	-0.57	-0.34	-0.32	-0.44
	alignment	+0.35	+0.53	+0.32	+0.21	+0.13
photo_z	global R^2	0.51	0.56	0.49	0.49	0.51
	shape	-0.06*	-0.05*	-0.17	+0.29	-0.16
	sphere	-0.33	-0.21	-0.29	-0.46	-0.44
	mismatch	-0.45	-0.61	-0.45	-0.48	-0.46
	alignment	+0.26	+0.33	+0.06*	+0.22	+0.14
smooth_fraction	global R^2	0.53	0.59	0.49	0.56	0.55
	shape	+0.07*	-0.08*	+0.20	+0.24	+0.06*
	sphere	-0.13	-0.06*	+0.05*	-0.15	-0.11
	mismatch	-0.09*	-0.33	-0.25	-0.20	-0.01*
	alignment	+0.19	-0.02*	+0.24	+0.04*	+0.14
stellar_mass	global R^2	0.48	0.54	0.50	0.49	0.49
	shape	+0.00*	-0.15	-0.10	-0.11	-0.13
	sphere	-0.35	-0.26	-0.38	-0.40	-0.38
	mismatch	-0.04*	+0.03*	-0.12	-0.06*	-0.02*
	alignment	+0.01*	-0.05*	-0.01*	+0.07*	-0.03*

Table 5: Cross-encoder replication, $d = 16$, multi-scale density control. * not significant at 0.05.

Appendix E: intervening on the readout's normal component

The partials of Table 1 compare anchors with one another and cannot say whether reversing the probe-facing shape term hurts relative to removing it, since no anchor is flat. Here we score a counterfactual local second-order surrogate of the readout at a fixed manifold. At each anchor the fitted ridge weight is split into tangent, radial and in-sphere normal parts $w = w_T + w_{\text{rad}} + w_S$; we score the anchor's 2,048 neighbours by the data-side tangent-plus-sphere readout $(w_T + w_{\text{rad}}) \cdot x$ plus tq with the local intercept refit, where $q = \frac{1}{2} \langle w_S, \Pi^S \rangle(u, u)$ is the decoder's in-sphere second-order term in the anchor's chart coordinates: $t = 1$ is the manifold's bending as seen in the readout's normal direction, $t = -1$ its sign reversal, $t = 0$ shape-flat (the sphere term $-(w \cdot \hat{x})g$ is kept in the base). Because the intercept is refit for every t , the local sum of squares is exactly $\text{SSE}(t) = \|r_0 - tq_c\|^2$ with r_0 the centred residual of the shape-flat predictor and $q_c = q - \bar{q}\mathbf{1}$ the centred quadratic on the anchor's neighbours; hence $t^* = \langle r_0, q_c \rangle / \|q_c\|^2$ and $t = 1$ beats shape-flat iff $2\langle r_0, q_c \rangle > \|q_c\|^2$. This is

encoder	label	mismatch, cross-fit		alignment, cross-fit		Hess. split cos p50
		fit A/score B	fit B/score A	fit A/score B	fit B/score A	
DINOv3	mag_r	−0.60	−0.62	+0.50	+0.50	+0.35
	photo_z	−0.56	−0.59	+0.28	+0.27	+0.24
	smooth_fraction	−0.32	−0.29	−0.03*	−0.02*	+0.31
	stellar_mass	+0.03*	+0.02*	−0.03*	−0.07*	+0.22
CLIP-B	mag_r	−0.28	−0.35	+0.28	+0.29	+0.31
	photo_z	−0.41	−0.40	+0.08*	+0.06*	+0.25
	smooth_fraction	−0.22	−0.24	+0.15	+0.20	+0.28
	stellar_mass	−0.14	−0.13	−0.04*	+0.01*	+0.19
ConvNeXt-B	mag_r	−0.36	−0.34	+0.19	+0.21	+0.29
	photo_z	−0.46	−0.45	+0.18	+0.19	+0.30
	smooth_fraction	−0.15	−0.19	−0.02*	+0.03*	+0.32
	stellar_mass	−0.05*	−0.01*	+0.14	−0.03*	+0.29
ViT-L	mag_r	−0.46	−0.49	+0.09*	+0.15	+0.30
	photo_z	−0.44	−0.42	+0.09*	+0.13	+0.24
	smooth_fraction	−0.03*	−0.04*	+0.11	+0.07*	+0.30
	stellar_mass	−0.06*	+0.01*	−0.04*	−0.03*	+0.21

Table 6: Cross-fitted mismatch and alignment partials per encoder (multi-scale control) and split-half reliability of the label Hessian.

253 the finite-sample counterpart of the tensor-level condition $2\langle R, K_S \rangle_F > \|K_S\|_F^2$ of Section 5, not
 254 the same quantity: the tensor version needs the estimated label Hessian, the empirical one does not.
 255 Columns: fraction of anchors where $t = 1$ beats shape-flat (help) and where $t = -1$ is worse than
 256 shape-flat (hurt), the median change in local R^2 at $t = \pm 1$, the median t^* ; then the same for a random
 257 in-sphere normal direction v whose quadratic is rescaled to the same centred amplitude on the actual
 258 neighbours, $\|q_{v,c}\| = \|q_c\|$ (matching $\|v\|$ to $\|w_S\|$ instead leaves the contracted tensor at a fraction
 259 of the fitted one, since Π^S spans at most $d(d+1)/2$ of the ~ 750 normal directions; matching
 260 $\|\langle v, \Pi^S \rangle\|_g$ gives the same picture; Supplement 12). The anchors’ neighbourhoods overlap (each
 261 point lies in about twelve of them), so the fractions are not counts of independent trials; a maximal
 262 set of anchors whose pairwise overlap is at most 5% of k has 19–21 members per run, on which
 263 $t = 1$ beats shape-flat at 81–100% (one-sided sign test $p \leq 10^{-3}$ in every cell) and $t = -1$ is worse
 264 at 95–100% ($p \leq 10^{-5}$).

run	label	decoder in-sphere shape term					random orientation, matched $\ q_c\ $				
		help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	t^*	help	hurt	$\Delta R^2(+1)$	$\Delta R^2(-1)$	
ViT-B, $d = 16$	mag_r	0.97	1.00	+0.030	-0.051	2.3	0.26	0.80	-0.009	-0.009	
	photo_z	0.99	0.99	+0.028	-0.038	3.0	0.33	0.68	-0.005	-0.006	
	smooth_fraction	0.90	1.00	+0.021	-0.040	1.7	0.24	0.78	-0.008	-0.010	
	stellar_mass	0.96	0.99	+0.026	-0.040	2.6	0.29	0.70	-0.006	-0.006	
ViT-B, $d = 20$	mag_r	0.99	1.00	+0.031	-0.049	2.6	0.24	0.78	-0.007	-0.008	
	photo_z	0.99	1.00	+0.029	-0.038	3.6	0.36	0.71	-0.004	-0.005	
	smooth_fraction	0.90	1.00	+0.020	-0.035	1.9	0.22	0.77	-0.007	-0.008	
	stellar_mass	0.97	1.00	+0.027	-0.038	3.1	0.32	0.72	-0.005	-0.005	
DINOv3, $d = 16$	mag_r	0.92	0.99	+0.018	-0.031	2.0	0.29	0.71	-0.007	-0.006	
	photo_z	0.96	0.99	+0.020	-0.030	2.6	0.29	0.71	-0.006	-0.004	
	smooth_fraction	0.93	1.00	+0.024	-0.055	1.5	0.22	0.78	-0.014	-0.013	
	stellar_mass	0.95	1.00	+0.016	-0.028	2.1	0.26	0.74	-0.005	-0.005	
CLIP-B, $d = 16$	mag_r	0.95	0.99	+0.018	-0.031	2.2	0.30	0.68	-0.006	-0.005	
	photo_z	0.99	1.00	+0.019	-0.026	3.4	0.36	0.64	-0.003	-0.003	
	smooth_fraction	0.92	0.99	+0.014	-0.027	1.7	0.25	0.69	-0.006	-0.005	
	stellar_mass	0.97	1.00	+0.020	-0.028	2.8	0.35	0.68	-0.004	-0.005	
ConvNeXt-B, $d = 16$	mag_r	0.92	1.00	+0.018	-0.031	2.0	0.26	0.73	-0.007	-0.007	
	photo_z	0.99	1.00	+0.018	-0.026	2.9	0.28	0.64	-0.004	-0.003	
	smooth_fraction	0.88	0.98	+0.016	-0.036	1.6	0.23	0.75	-0.009	-0.008	
	stellar_mass	0.99	1.00	+0.022	-0.032	3.1	0.29	0.65	-0.005	-0.003	
ViT-L, $d = 16$	mag_r	0.96	0.99	+0.022	-0.034	2.6	0.32	0.74	-0.005	-0.006	
	photo_z	0.97	0.99	+0.018	-0.025	3.2	0.34	0.67	-0.004	-0.003	
	smooth_fraction	0.79	0.97	+0.013	-0.029	1.6	0.25	0.77	-0.006	-0.008	
	stellar_mass	0.98	1.00	+0.022	-0.032	2.9	0.30	0.67	-0.004	-0.004	

Table 7: Counterfactual scaling of the in-sphere shape term in a local second-order surrogate of the readout, 512 anchors per run. The manifold’s bending as seen in the readout’s normal direction helps at nearly every anchor, its sign reversal hurts at nearly every anchor, a random orientation of the same centred quadratic amplitude shows no such asymmetry and is typically harmful under either sign; $t^* > 1$ throughout, consistent with the globally ridge-regularized probe under-using a locally beneficial term.